Robust Automated Spelling Correction with Deep Ensembles

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ABSTRACT

Spelling correction is one of the most elusive problems in NLP, especially for agglutinative languages, such as Azerbaijani. Rich morphology of these languages makes statistical methods less effective. There have been several attempts to use neural networks to solve the problem. Traditional approach consists of autoregressive encoder-decoder models. However, they suffer from the problem of uncertain predictions, where originally correct words are regenerated incorrectly. This affects model's utility negatively, since unreliable models are avoided in production-level systems. In this paper, we assess the problem of uncertainty, and attempt to solve this problem by modifying the original architecture to get more robust predictions. We have tried out several methods, including a deep ensemble and carried out comparative analysis.

CCS CONCEPTS

• **Applied computing** → **Document management and text processing**; • **Computing methodologies** → **Machine learning algorithms**.

KEYWORDS

spelling correction, uncertainty estimation, neural networks, deep ensemble learning

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1 INTRODUCTION

1.1 Spelling correction

High ratio of spelling mistakes in a text can render it incomprehensible to humans and machines alike. Automated spelling correction is not only a matter of convenience for the readers. It is also a highly integrated part of text processing pipelines in most cases. This integration ensures improved performance on various NLP tasks, such as sentiment analysis and named entity recognition. Spelling errors, once detected, can also be used as a new feature, which can further boost predictive capabilities of the machine learning models.

1.2 Azerbaijani language

English language has robust solutions for the problem of spelling correction. However, existing solutions for Azerbaijani language are lacking in several aspects. While there are some general explanations for this, a major one being the fact that there are less resources for this language, we also need to consider language-specific factors. Agglutinative nature of the Azerbaijani language renders statistical methods ineffective. [9] goes into detail on challenges of spelling correction for agglutinative languages. Deep learning solutions have been shown to be more effective in the case of Azerbaijani, as well as other agglutinative languages. However, this solution comes with its own set of problems. Conventional deep learning models have no proper means of estimating uncertainty. This makes deep learning solutions more prone to errors in out-of-distribution inputs. In the case of generative models, this results in hallucination.

There are certain cognitive errors that are peculiar to Azerbaijani. There is an interesting case that emerged as a result of multiple keyboards being used in the country. Some people use the traditional QWERTY keyboard used across the globe, while others have begun to use the QÜERTY keyboard that includes additional letters in Azerbaijani. While the latter, modern keyboards are sufficient to write correctly in Azerbaijani. Widespread use of the English keyboards has resulted in accumulation of incorrect text corpora. For example, "Azərbaycan" is written as "Azerbaycan," "dərə" is written as "dere," etc. Another class of cognitive mistakes includes misspellings due to the difference between spoken and written forms of the word. For example, "onundur" can be written as "onundu" in informal setting. Yet another class of cognitive mistakes emerge as a result of lack of grammatical knowledge. For example, "mənim də" is often written as "mənimdə."

1.3 Proposal

Neural networks with encoder-decoder architecture have long been used for machine translation [6]. We propose to modify this model in several ways that will allow us to assess certainty of the model's output.

One proposition is to directly use output of the final softmax layer in an uncertainty estimation process. Another proposition involves an ensemble of the base LSTM models. While the LSTM layer ensures that the model will learn the relationship between

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the letters, using an ensemble instead of a single model helps with the uncertainty estimation.

Our contributions are as follows:

- We build a deep learning model for spelling correction in Azerbaijani
- We design and evaluate three methods for uncertainty estimation in neural spelling correction
- As a result, we address the problem of maintaining originally correct words, which has not received enough attention in previous works.

Paper is structured as follows: Next section gives a brief overview of the previous works related to spelling correction and uncertainty estimation. After this, we provide a detailed explanation of the core idea. This is followed by experiments and their results.

2 LITERATURE REVIEW

Error detection can be defined as a subproblem of spelling correction – wrong words have to be detected before we attempt to fix them. There is substantial literature on this particular problem. The most common approach to this problem involves machine learning algorithms. The problem is reframed as a binary classification problem, where two classes are "correct" and "misspelt." A dataset is curated with these labels. And a machine learning model is trained on it. [2] uses an advanced version of this. They use two deep learning models. First model detects incorrect words, and the second model identifies false positives.

[10] uses Levenshtein and Damerau-Levenshtein distances as spelling error metric. More novel approach to spelling correction is re-framing the problem as a translation problem (from incorrect to correct word). This allows us to use sequence-to-sequence (Seq2Seq) models. This approach has been implemented for Azerbaijani language with some success by [8]. [4] has made a more successful attempt for the Chinese language by uses transformers architecture.

An alternative solution includes non-deterministic finite state machines. This method involves dividing words into parts and predicting the word by walking through a state machine to build the whole word. It is claimed to work particularly well for agglutinative languages. [5] applies this approach to the Kurdish language.

Yet another approach to spelling correction involves adding information about the word other than its writing. In the case of Chinese spelling correction, this translates to integrating pinyin of the word to the model input. [13] provides a prime example of this approach.

Contextual spelling correction solutions take as input not only the word but also its context. This gives the model a broader understanding of the word, and higher likelihood of an accurate correction. [3] uses n-grams to take advantage of the context in spelling correction. [12] uses deep learning techniques to create a contextual spelling correction model for Automatic Speech Recognition systems.

3 METHODS

3.1 Dataset

Before building any deep learning models, we need a dataset of certain quality. It needs to contain both original (possibly incorrect) and correct spellings. The latter is going to serve as labels during the training. This corpus needs to be:

- Clean, i.e. minimal unnecessary content,
- Variety in spelling mistakes
- Variety in words

Social media content provides immense variety in both content and misspellings. Cleanliness can be achieved with arduous processing. We scraped social media posts and comments in Azerbaijani. We extracted words from this corpus, and used a team of manual annotators to label the words. In the end, we had 48866 labelled words. 45% percent of these words were originally correct. That's to say, the Levenshtein distance between the original word and the label was 0. Dataset contained 84 unique characters. This includes upper and lower-case letters, numbers, and punctuation marks. Labels contained 82 unique characters. Most of the mistakes were result of using English keyboard instead of Azerbaijani. Azerbaijani alphabet contains 32 letters, so certain letters are often replaced with their closest English equivalent. For example, Ə - E, Ö - O, etc. Other common mistakes include combining certain particles to words (e.g., "onun da" instead of "onunda").

We initially split the dataset into train and test parts, with 20% percent being the test dataset. Train dataset was further divided into train and validation datasets to be used during training, again with 4/1 ratio.

3.2 Models

We have devised three different approaches:

3.2.1 Base model. Our base model has a simple architecture. One LSTM layer functions as the encoder. Decoder consists of an LSTM and a dense layer. LSTM is an advanced form of recurrent neural layer that deals with diminishing gradient problem. Compared to RNNs, LSTM models are capable of learning longer sequences [7].

During the training process, both the correct and incorrect words are fed to the models after vectorisation. During the inference, original words are fed through the encoder, and decoder generates the prediction at character-level, in an autoregressive manner.

This method is going to be referred to as the **base method** through the paper.

3.2.2 Delta and entropy models. Our second solution also relies on this model, but has a different final decision algorithm. While the predicted word is being generated, delta D is calculated for each generated letter, including the stop token. D is defined as the difference between the first highest and second highest values in the output vector of the final (softmax) layer. In the end the average of these deltas (D_{avg}) is calculated. If D_{avg} is below a certain threshold T, then the prediction is rejected and the original word is provided as the output. This method is going to be referred to as the **delta method** through the paper.

A variation of this method involves entropy function instead of simple subtraction. In this case, everything is the same as the

delta method, except for one thing. Instead of calculating the difference between first and second highest softmax scores, we treat the vector as a distribution and calculate Shannon's entropy [11] of it. Shannon's entropy attempts to capture the inherent uncertainty in a distribution. This method is going to be referred to as the **entropy** method through the paper.

3.2.3 Ensemble model. Third approach consists of deep ensembles. We trained N models, all of them with the same architecture, data, and hyperparameters. The only difference between the models becomes the seed used to initialise the weights. Deep ensembles have been successfully used to estimate uncertainty elsewhere, but we are not aware of its application to spelling correction. Figure 1 shows the idea briefly. Decision algorithm works as follows: It takes an input K. If at least K models out of N predict the same string, this is accepted as correct and provided as the output. Otherwise, all predictions are rejected and the original word itself becomes the output. This method is going to be referred to as the **ensemble** method through the paper.

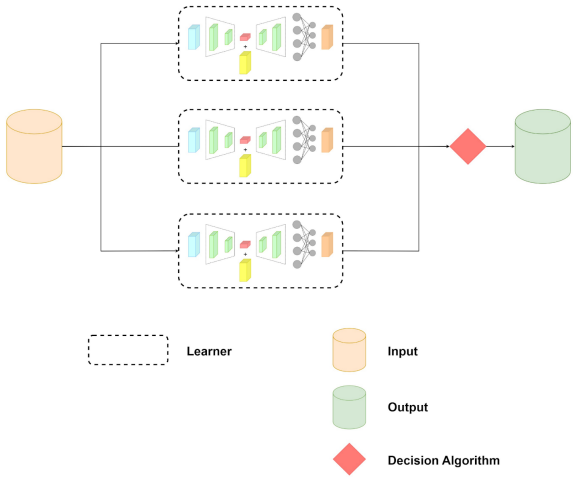


Figure 1: Ensemble method.

4 EXPERIMENTS

We have performed a set of experiments to assess the performance of the aforementioned methods against each other, as well as against earlier works. Experiments consist of four parts: (1) base experiments to establish our baseline, (2) delta and (3) entropy experiments show our efforts to handle uncertainty with a single model, and (4) ensemble experiments to evaluate our final solution. Codebase is publicly available at the following URL:

https://github.com/ceferisbarov/spelling-correction-tf

4.1 Experimental setup

Our models have been trained on NVIDIA GeForce RTX 3060. Inference has been performed on Intel® Core™ i7-11800H Processor. All of the code has been developed and run on Ubuntu 22.04. We have used TensorFlow 2.10 [1] and Python 3.10 for modelling.

8 base models were trained. All of the experiments have been performed on these models. No separate model training was necessary for delta, entropy or ensemble methods. These 8 models have the same architecture, were trained on the full dataset. Their weights were initialized with different seeds. They were trained for 20 epochs, with batch size of 32. We used Tensorflow's Adam optimizer with default configurations. Loss function was categorical cross-entropy. Post-training quantization has been performed on all models before evaluation.

4.2 Base experiments

First experiment evaluates performance of base models as-is. 8 base models are evaluated on the test dataset and the variation in their performance is recorded. This variation is explained only via the different weight initialization points. We use word error rate (WER) through the rest of the experiments. You can see the results in Table 1.

Table 1: Results of experiments per model (word error rate).

Model	Base	Delta	Entropy
1	27.00	24.90	24.76
2	27.62	25.35	25.37
3	27.04	25.39	25.05
4	28.83	26.01	25.72
5	26.07	23.71	23.68
6	28.18	25.20	25.16
7	25.78	24.12	24.03
8	25.38	23.57	23.41

By separately assessing the performance of the model on originally correct and incorrect words, we can observe that a considerable portion of the originally correct words are misspelled by the model. You can see the comparison in Figure 4.

4.3 Delta experiments

Second set of experiments evaluate the performance of the delta method.

Initial experiment was performed to find the ideal threshold for the delta method. As we can see in Figure 2, the best threshold is 0.85, where WER is the lowest.

Later, all 8 base models were evaluated with this decision algorithm. Performance has been increased by roughly 1-1.5% across all models, with slight variation. See Table 1 for further details.

When looking at the performance difference between originally correct and incorrect words, we can observe that originally correct words are retained better than the base model (See Figure 4). There is no significant change in the second (originally incorrect) group.

4.4 Entropy experiments

Third set of experiments evaluate the performance of the entropy method in a similar fashion.

First, we identify the ideal threshold. You can these results in Figure 1. Then we evaluate all base models using the test dataset.

Results of the entropy method are only marginally better than the delta model. As we can see in Table 1, comparison of originally

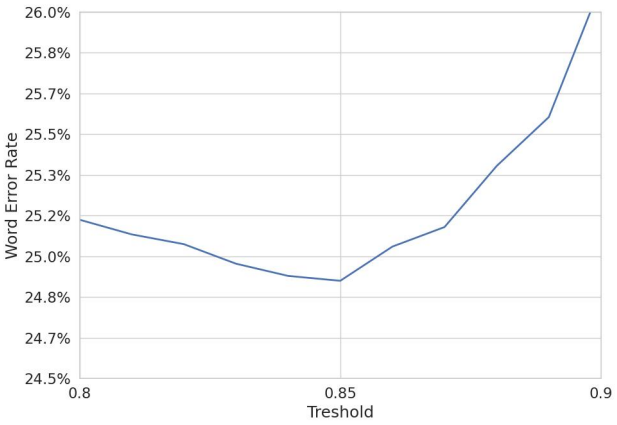


Figure 2: Performance of the delta method with various thresholds.

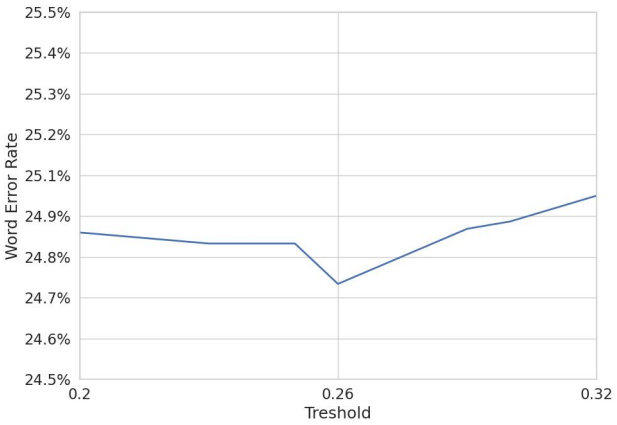


Figure 3: Performance of the entropy method with various thresholds.

correct and incorrect words gives a similar result to that of the delta.

4.5 Ensemble experiments

Fourth and final set of experiments evaluate ensemble method. At first, we optimize two hyperparameters – N, total number of models in the ensemble, and K, the threshold. Table 2 shows WER across different combinations of these two hyperparameters. Note that $K = 1$ has been ignored, since it is essentially equal to only using the base model. The best performing combination is ($N = 8, K = 3$).

4.6 Final results

Results of four experiments have been summarized in Table 3. Ensemble method has the best performance both in terms of WER and CER. Delta method improves performance over the base model by 1-1.5%. Entropy method provides only marginal improvement upon the delta method.

Table 2: Ensemble experiment results (WER)

.	2	3	4	5	6	7	8
2	26.14	-	-	-	-	-	-
3	20.04	29.54	-	-	-	-	-
4	18.80	22.25	31.29	-	-	-	-
5	18.45	19.56	24.13	33.07	-	-	-
6	18.33	18.36	21.16	25.73	34.67	-	-
7	18.34	18.00	19.15	22.57	27.35	35.92	-
8	18.34	**17.67**	18.08	20.36	23.67	28.44	36.71

Table 3: Final results

Method	Threshold	Mean WER	Mean CER
Base	-	25.38	6.32
Delta	0.85	23.57	5.24
Entropy	0.26	23.41	5.18
Ensemble	3/8	**17.67**	**3.64**

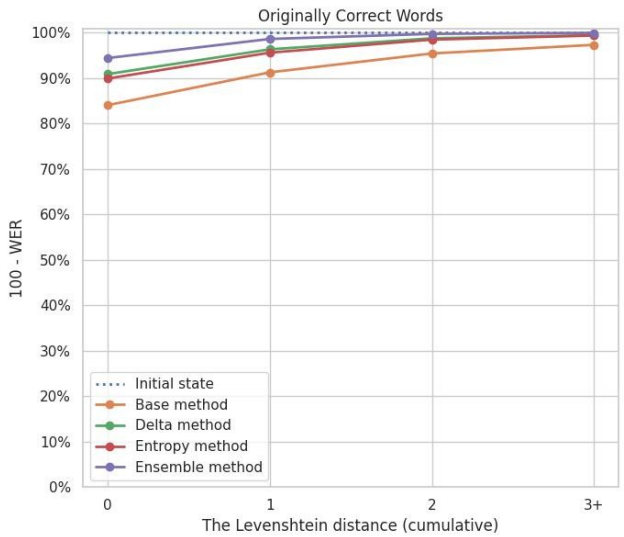
Evaluating the originally correct and incorrect words separately provides a clearer picture. Ensemble method performs the best in both cases (See Figure 4). Delta and Entropy methods show some improvement on originally correct words. That means they are better than the base model at retaining correct words during the generation process. However, they show similar performance to the base model when tested on originally incorrect words.

5 CONCLUSION

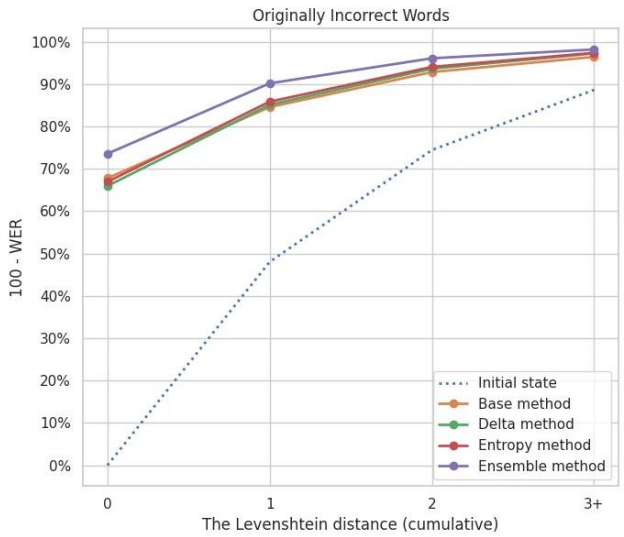
In this paper, we presented the problem of uncertainty estimation in spelling correction, suggested several solutions, implemented them and evaluated their performance to find the best approach. Initial method, which involves an LSTM-based autoregressive neural network, suffers from uncertain predictions. We attempted to solve this problem via three different methods: delta, entropy, and ensemble. Delta and entropy methods utilize the output of the softmax function in different ways to assess certainty of the predictions, while ensemble method has a multi-model approach. It consists of multiple models with same architecture and training data but different weight initialization seeds. Ensemble method outperformed all other alternatives.

This work can be expanded in two ways:

(1) Directly using output of the softmax function is not the best way to measure uncertainty. Temperature scaling can be used to calibrate the scores to this end.
(2) These methods have only been tried on Azerbaijani. It would be interesting to see how these results hold in other languages.
(3) Systemic evaluation can be performed to assess the performance against other established methods.



(a)



(b)

Figure 4: (a) Performance of the four methods on originally correct part of the dataset. (b) Performance on originally incorrect (Levenshtein distance > 1) words.

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